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Anode holder, and plating apparatus

Abstract

Provided are an anode holder capable of reducing consumption of an additive in a plating apparatus, and a plating apparatus. An anode holder for holding an anode for use in a plating apparatus is provided, and the anode holder includes an inner space formed in the anode holder, to house the anode, a mask including a plurality of holes, and configured to cover a front surface of the inner space, and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers the front surface of the inner space.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an anode holder and a plating apparatus.

BACKGROUND ART

(2) Heretofore, a wiring has been formed in a fine wiring groove, hole or resist opening provided in a surface of a semiconductor wafer or the like, and a bump (a protruding electrode) to be electrically connected to an electrode of a package or the like has been formed on the surface of the semiconductor wafer or the like. As a method of forming this wiring and bump, for example, an electroplating method, an evaporation method, a printing method, a ball bump method or the like is known, but with increase in an I/O number of semiconductor chips and for a finer pitch, the electroplating method is becoming often used in which miniaturization is possible and performance is relatively stable.

(3) A plating apparatus for use in the electroplating method includes a substrate holder holding a substrate of a semiconductor wafer or the like, an anode holder holding an anode, and a plating solution tank that stores a plating solution containing a large number of types of additives. When a substrate surface of the semiconductor wafer or the like is plated in this plating apparatus, the substrate holder is disposed to face the anode holder in the plating solution tank. In this state, the substrate and the anode are energized, and accordingly a plating film is formed on the substrate surface. In addition, the additive has an effect of accelerating or suppressing a film formation speed of the plating film, an effect of improving film quality of the plating film, and the like.

(4) Heretofore, a soluble anode that dissolves in the plating solution or an insoluble anode that does not dissolve in the plating solution has been used as the anode to be held by the anode holder. In a case where the plating is performed by using the insoluble anode, oxygen is generated by reaction between the anode and the plating solution. The additive in the plating solution reacts with this oxygen and is decomposed. There is a problem that, when the additive is decomposed, the additive loses the above-described effects, and a desired film cannot be obtained on the substrate surface (e.g., see PTL 1). It is also known that, in a case where, for example, phosphorus-containing copper is used as the soluble anode, deterioration of the additive, especially an accelerator, occurs due to reaction with monovalent copper generated from the anode when electrolysis is not performed. To prevent this problem, the additive may be added to the plating solution as required to keep concentration of the additive in the plating solution in a predetermined concentration or more. However, the additive is expensive, and hence it is desirable to inhibit the decomposition of the additive as much as possible.

(5) Consequently, it has been suggested that an interior of the plating solution tank is divided, by a diaphragm, into a space in which the anode is disposed (an anode tank) and a space in which the substrate and a cathode are arranged (a cathode tank), to inhibit the additive in the plating solution from reaching the anode and suppress decomposition of the additive (e.g., see PTL 2).

CITATION LIST

Patent Literature

(6) PTL 1: Japanese Patent No. 2510422 PTL 2: Japanese Patent Laid-Open No. 2009-155726

SUMMARY OF INVENTION

Technical Problem

(7) As described above, by a diaphragm including a fine hole with a size smaller than an average size of a molecule included in an additive, an additive included in a plating solution in a cathode tank is inhibited from being moved into an anode tank, and decomposition of the additive is suppressed. Heretofore, the diaphragm has been disposed to cover an opening in an anode holder, an anode box, or a regulation plate. However, according to investigation by the present inventors, it has been found that in a conventional configuration, a region where the diaphragm acts on the plating solution in the cathode tank is wide, the additive is therefore consumed, and there is room for improvement.

(8) The present invention has been developed in view of the above problems, and one of objects thereof is to provide an anode holder capable of reducing consumption of an additive in a plating apparatus, and a plating apparatus.

Solution to Problem

(9) According to an embodiment of the present invention, an anode holder for holding an anode for use in a plating apparatus is provided, and the anode holder includes an inner space formed in the anode holder, to house the anode, a mask including a plurality of holes, and configured to cover a front surface of the inner space, and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers the front surface of the inner space. According to this anode holder, the mask can reduce a region where the diaphragm is in contact with a plating solution, and can further inhibit the additive from reaching the anode to reduce consumption of the additive.

(10) According to another embodiment of the present invention, a plating apparatus is provided, and the plating apparatus includes a plating solution tank, a mask including a plurality of holes, and dividing the plating solution tank into an anode tank in which an anode is disposed and a cathode tank in which a cathode is disposed, and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers a front surface of an inner space. According to this plating apparatus, the mask can reduce a region where the diaphragm is in contact with a plating solution, and can further inhibit an additive from reaching the anode to reduce consumption of the additive.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view showing a plating apparatus according to a first embodiment;
- (2) FIG. 2 is a plan view of an anode holder according to the present embodiment;
- (3) FIG. 3 is a side cross-sectional view of an anode holder **60** taken along the 3-3 line shown in FIG. 2;
- (4) FIG. 4 is an exploded perspective view of the anode holder in a state where a holder base cover is removed;
- (5) FIG. 5 is a plan view of the anode holder in the state where the holder base cover is removed;
- (6) FIG. 6A is a view schematically showing a mounting structure of a diaphragm and a mask in FIG. 3;
- (7) FIG. 6B is a view schematically showing another example of the mounting structure of the mask shown in FIG. 6A;
- (8) FIG. 7 is a view schematically showing a mounting structure of a diaphragm and a mask according to a first modification;
- (9) FIG. 8 is a view schematically showing a mounting structure of a diaphragm and a mask according to a second modification;
- (10) FIG. 9 is a view schematically showing a mounting structure of a diaphragm and a mask according to a third modification;
- (11) FIG. 10 is a view schematically showing a mounting structure of a diaphragm and a mask according to a fourth modification;
- (12) FIG. 11 is a view schematically showing a mounting structure of a diaphragm and a mask according to a fifth modification;
- (13) FIG. 12 is a view schematically showing a mounting structure of a diaphragm and a mask according to a sixth modification;
- (14) FIG. 13 shows a fixed part between a diaphragm and a mask according to a first example;
- (15) FIG. 14 shows a fixed part between a diaphragm and a mask according to a second example;

(16) FIG. 15 shows a fixed part between a diaphragm and a mask according to a third example;
(17) FIG. 16 shows a fixed part between a diaphragm and a mask according to a fourth example;
(18) FIG. 17 shows a fixed part between a diaphragm and a mask according to a fifth example;
(19) FIG. 18 shows a fixed part between a diaphragm and a mask according to a sixth example;
(20) FIG. 19 shows a fixed part between a diaphragm and a mask according to a seventh example;
(21) FIG. 20 shows a fixed part between a diaphragm and a mask according to an eighth example;
(22) FIG. 21 shows a fixed part between a diaphragm and a mask according to a ninth example;
(23) FIG. 22 is a schematic view showing a plating apparatus according to a second embodiment;
and

(24) FIG. 23 is a schematic view showing a plating apparatus according to a third embodiment.

DESCRIPTION OF EMBODIMENTS

(25) Hereinafter, embodiments of a plating apparatus and an anode holder according to the present invention will be described with reference to the accompanying drawings. In the accompanying drawings, the same or similar element is denoted with the same or similar reference sign, and in descriptions of the respective embodiments, a description concerning the same or similar element may not be repeated. Also, characteristics illustrated in the respective embodiments are also applicable to another embodiment as long as the characteristics of the embodiments are not contradictory to each other.

First Embodiment

(26) FIG. 1 is a schematic view showing a plating apparatus according to a first embodiment. As shown in FIG. 1, the plating apparatus includes a plating solution tank 50 holding a plating solution inside, an anode 40 disposed in the plating solution tank 50, an anode holder 60 holding the anode 40, and a substrate holder 18. The substrate holder 18 removably holds a substrate W such as a wafer, and is configured to immerse the substrate W into the plating solution in the plating solution tank 50. The plating apparatus according to the present embodiment is an electroplating apparatus that applies current through the plating solution to plate a surface of the substrate W with a metal.

(27) The substrate W is, for example, a semiconductor substrate, a glass substrate, or a resin substrate. The metal with which the surface of the substrate W is plated is, for example, copper (Cu), nickel (Ni), tin (Sn), Sn—Ag alloy, or cobalt (Co).

(28) The anode 40 and the substrate W are arranged to extend in a vertical direction, that is, so that plate surfaces of the anode 40 and the substrate W face in a horizontal direction and face each other in the plating solution. The anode 40 is connected to a positive electrode of a power source 90 via the anode holder 60, and the substrate W is connected to a negative electrode of the power source 90 via the substrate holder 18. When a voltage is applied between the anode 40 and the substrate W, current flows to the substrate W, and a metal film is formed on the surface of the substrate W in the presence of the plating solution.

(29) The plating solution tank 50 includes a plating solution storage tank 52 in which the substrate W and the anode 40 are arranged, and an overflow tank 54 disposed adjacent to the plating solution storage tank 52. The plating solution in the plating solution storage tank 52 flows over a side wall of the plating solution storage tank 52 to flow into the overflow tank 54.

(30) One end of a plating solution circulation line 58a is connected to a bottom of the overflow tank 54, and the other end of the plating solution circulation line 58a is connected to a bottom of the plating solution storage tank 52. A circulation pump 58b, a constant temperature unit 58c and a filter 58d are attached to the plating solution circulation line 58a. The plating solution flows over the side wall of the plating solution storage tank 52 to flow into the overflow tank 54, and further flows from the overflow tank 54 through the plating solution circulation line 58a to return to the plating solution storage tank 52. Thus, the plating solution circulates between the plating solution storage tank 52 and the overflow tank 54 through the plating solution circulation line 58a.

(31) The plating apparatus further includes a regulation plate 14 that regulates a potential distribution on the substrate W, and a paddle 16 that stirs the plating solution in the plating solution

storage tank 52. The regulation plate 14 is disposed between the paddle 16 and the anode 40, and includes an opening 14a for limiting an electric field in the plating solution. The paddle 16 is disposed in the vicinity of the surface of the substrate W held by the substrate holder 18 in the plating solution storage tank 52. The paddle 16 is made of, for example, titanium (Ti) or resin. The paddle 16 reciprocates in parallel with the surface of the substrate W, to stir the plating solution so that metal ions are sufficiently and uniformly supplied to the surface of the substrate W during the plating of the substrate W.

(32) FIG. 2 is a plan view of the anode holder 60, FIG. 3 is a side cross-sectional view of the anode holder 60 taken along the 3-3 line shown in FIG. 2, FIG. 4 is an exploded perspective view of the anode holder 60 in a state where a holder base cover 63 is removed, and FIG. 5 is a plan view of the anode holder 60 in the state where the holder base cover 63 is removed. Note that FIG. 5 shows, for convenience, the anode holder 60 in a state where a grip 64-2 is transparent.

Furthermore, FIGS. 4 and 5 show, for convenience, the anode holder 60 in a state where the anode 40 is removed. Further, in the present description, “up” and “down” refer to an upward direction and a downward direction in a state where the anode holder 60 is vertically housed in the plating solution tank 50. Similarly, in the present description, “a front surface” refers to a surface on a side on which the anode holder 60 faces the substrate holder, and “a back surface” refers to a surface on a side opposite to the front surface.

(33) As shown in FIGS. 2 to 4, the anode holder 60 according to the present embodiment includes a substantially rectangular holder base 62 including an inner space 61 that houses the anode 40, a pair of grips 64-1 and 64-2 formed in an upper part of the holder base 62, and a pair of arms 70-1 and 70-2 similarly formed in the upper part of the holder base 62. Also, the anode holder 60 includes the holder base cover 63 that partially covers a front surface of the holder base 62, a diaphragm 66 disposed on a front surface of the holder base cover 63 to cover the inner space 61, a mask 67 including a plurality of holes 67a and fixed to the diaphragm 66, and an outer edge mask 68 disposed on front surfaces of the diaphragm 66 and the mask 67. Additionally, in the present embodiment, the holder base cover 63 supporting the diaphragm 66 and the mask 67 corresponds to “a base body”.

(34) As shown in FIGS. 2 and 5, the holder base 62 includes a hole 71 extending from an outer surface of a lower part to the inner space 61 of the holder base, and communicating with the inner space 61. Also, the holder base 62 includes an air outlet 81 for exhausting air of the inner space 61, between the grips 64-1 and 64-2 in the upper part of the holder base. When the holder base 62 is immersed into the plating solution, the plating solution flows through the hole 71 into the inner space 61, and air of the inner space 61 is exhausted from the air outlet 81. Furthermore, in a case where the insoluble anode is used as the anode 40, oxygen generated from the anode 40 during the plating is also exhausted through the air outlet 81. The air outlet 81 is closed with a lid 83 formed so that the exhaust of air is not obstructed.

(35) Also, as shown in FIG. 3, an annular opening 63a having a diameter larger than a diameter of the anode 40 is formed in a substantially central portion of the holder base cover (base body) 63. The holder base cover 63 forms the inner space 61 together with the holder base 62. The diaphragm 66 is disposed on a front surface of the opening 63a, to close the inner space 61. In the present embodiment, the mask 67 including the plurality of holes 67a is fixed to one plate surface of the diaphragm 66. A diaphragm retainer 69 is attached in front of outer peripheral edges of the diaphragm 66 and the mask 67, and the outer edge mask 68 is disposed in front of the diaphragm retainer 69. Also, an annular first sealing member 84 including, for example, an O-ring or the like is disposed along the opening 63a in the front surface of the holder base cover 63. The diaphragm 66 and the mask 67 are pressed onto the first sealing member 84 by the diaphragm retainer 69, to tightly close the opening 63a. That is, the first sealing member 84 can tightly close between the diaphragm 66 and the inner space 61. Consequently, the inner space 61 and the outer space are divided by the diaphragm 66 and the mask 67.

(36) The diaphragm **66** is, for example, an ion exchange membrane such as a cation exchange membrane, or a neutral diaphragm. During plating, cations can pass through the diaphragm **66** from an anode side to a cathode side while any additive in the plating solution does not pass. A specific example of the diaphragm **66** is YUMICRON (registered trademark) manufactured by Yuasa Membrane Systems Co., Ltd.

(37) The mask **67** is a plate-shaped member including the plurality of holes **67a**, and is disposed to reduce a region where the diaphragm **66** is in contact with the plating solution. The mask **67** has a plate thickness of, for example, about 1 mm. The mask **67** is made of a resin such as polypropylene (PP) or polyvinyl chloride (PVC), a metal such as titanium (Ti) or the like. The mask **67** is fixed to the plate surface of the diaphragm **66**. In an example shown in FIGS. 2 to 5, the mask **67** is fixed to a front of the diaphragm **66**, that is, on an outer space side (side opposite to the inner space **61**) of the diaphragm **66**. However, the present invention is not limited to this example, and the mask **67** may be fixed to a rear of the diaphragm **66**, that is, on an inner space **61** side of the diaphragm **66**, and may be fixed to both the front and the rear of the diaphragm **66**.

(38) In the mask **67**, the plurality of holes **67a** are formed. In each of the plurality of holes **67a**, for example, a maximum distance from one end to the other end (an inner diameter in a case where the plurality of holes **67a** each have a circular shape) is preferably 10 mm or less, especially preferably 8 mm or less, 5 mm or less, 3 mm or less, or 2 mm or less. Also, it is preferable that each of the plurality of holes **67a** has the circular shape, but may have an elliptic shape, a polygonal shape or the like. Furthermore, in the example shown in FIGS. 2 to 5, each of the plurality of holes **67a** has the same size, but the present invention is not limited to this example. For example, the plurality of holes **67a** may have a size increasing closer to a center of the anode **40**, and decreasing away from the center of the anode **40**, and conversely, the holes may have a size decreasing closer to the center of the anode **40**, and increasing away from the center of the anode **40**. Also, in the example shown in FIGS. 2 to 5, the plurality of holes **67a** are provided at equal intervals in a biaxial direction in a plate surface of the mask **67**, but the present invention is not limited to this example. For example, the plurality of holes **67a** may be arranged with a distance between the holes decreasing closer to the center of the anode **40** and increasing away from the center of the anode **40**, and conversely the holes may be arranged with the distance between the holes increasing closer to the center of the anode **40** and decreasing away from the center of the anode **40**. Furthermore, the plurality of holes **67a** may be radially arranged.

(39) Also, in the plurality of holes **67a**, it is preferable that an opening ratio is equal to or more than 2% and equal to or less than 25%, and it is especially preferable that the opening ratio is equal to or more than 3%, equal to or more than 5%, equal to or less than 10%, or equal to or less than 12.5%. This is based on the fact that, if the opening ratio is large, the contact region of the diaphragm **66** with the plating solution is large, an effect of reducing consumption of an additive accordingly decreases, and it is difficult to sufficiently fix the diaphragm **66** and the mask **67**. This is also based on the fact that, if the opening ratio is small, it is difficult to remove gas (bubble) from the holes **67a**, and passage of cations from the anode side to the cathode side through the diaphragm **66** runs short. In the present embodiment, the plurality of holes **67a** are substantially uniformly arranged, and the opening ratio due to the plurality of holes **67a** is 6%. However, the present invention is not limited to this example. For example, the mask **67** may be formed with the opening ratio decreasing closer to the center of the anode **40** and increasing away from the center of the anode **40**, and conversely, the mask may be formed with the opening ratio increasing closer to the center of the anode **40** and decreasing away from the center of the anode **40**.

(40) Further, the plurality of holes **67a** may be formed with the same diameter in a front-rear direction, or may be formed to be tapered. It is especially preferable that the plurality of holes **67a** of the mask **67** may be formed to be tapered with a diameter decreasing closer to the diaphragm **66** and increasing away from the diaphragm **66**. In this case, foreign matter such as gas or bubbles can be inhibited from staying in the holes **67a**.

(41) The mask **67** is fixed to the diaphragm **66**. In other words, the diaphragm **66** is fixed to the mask **67**. At least part of the diaphragm **66** is fixed to the mask **67** in a region of the mask **67** that covers a front surface of the inner space **61**, that is, a region of the mask that covers the opening **63a** of the holder base cover **63**. However, the diaphragm **66** and the mask **67** may be fixed to each other also in a region other than the region that covers the front surface of the inner space **61**. In addition, it may be also considered that the mask **67** is “secured” to the diaphragm **66**.

(42) In the present embodiment, the mask **67** is attached to the diaphragm **66** by welding. However, a method of fixing the mask **67** to the diaphragm **66** is not limited to the welding. For example, the diaphragm **66** and the mask **67** may be non-removably welded, pressed or bonded (hereinafter, referred to together as “closely connected”) via a closely connecting layer. Specifically, the diaphragm **66** and the mask **67** may be closely connected to each other by heat welding with a sealer or the like, laser welding, ultrasonic welding, or vibration welding. Alternatively, the diaphragm **66** and the mask **67** may be closely connected to each other by using a pouch processing technology, a laminate processing technology, or an adhesive such as vinyl chloride. In the pouch processing technology and the laminate processing technology, attaching a sheet material such as a PET material at high temperature and high pressure, attaching sheet materials such as PET materials to each other by plasma treatment, or extruded lamination by use of the sheet material such as a PET material may be adopted. Alternatively, as the adhesive, TAKIBOND (registered trademark) that is an adhesive for PVC manufactured by TAKIRON Corporation, an epoxy resin adhesive for PE and PET, or a low outgas adhesive manufactured by Sunstar Engineering may be adopted.

(43) The mask **67** and the diaphragm **66** may be non-removably closely connected in the whole region of the mask **67**, or may be non-removably closely connected to be fixed to each other in part of the region. However, the plating solution enters a gap between the mask **67** and the diaphragm **66** to increase a contact region of the diaphragm **66** with the plating solution. Especially in the plating apparatus of the present embodiment, the plating solution is stirred with the paddle **16**, and hence the plating solution easily enters the gap between the mask **67** and the diaphragm **66**. Consequently, it is preferable that the mask **67** and the diaphragm **66** are non-removably closely connected in a wide region to reduce entrance of the plating solution into the gap.

(44) Thus, the anode holder **60** of the present embodiment includes the mask **67** including the plurality of holes **67a** and covering the front surface of the inner space **61**, and the diaphragm **66** is disposed to be fixed to the mask **67**. Consequently, the region where the diaphragm **66** is in contact with the plating solution can be smaller than that in a case where the mask **67** is not provided, and the additive can be inhibited from reaching the anode **40** to reduce the consumption of the additive.

(45) The outer edge mask **68** is a plate-shaped member including an annular opening in a central portion of the member, and is removably mounted to a front surface of the diaphragm retainer **69**. A diameter of the opening in the outer edge mask **68** is smaller than an outer diameter of the anode **40**. Consequently, when the outer edge mask **68** is attached to the diaphragm retainer **69**, the outer edge mask **68** is configured to cover an outer peripheral edge of the anode **40** when seen from a plane shown in FIG. 2. Consequently, the outer edge mask **68** can control an electric field on the surface of the anode **40** during the plating.

(46) The holder base cover **63** is fixed to the holder base **62** by screw connection, welding or the like, and the holder base cover **63** is closely connected to the holder base **62**. Alternatively, the holder base cover **63** may be formed integrally with the holder base **62**.

(47) As shown in FIGS. 2, 4 and 5, the grips **64-1** and **64-2** are coupled with the holder base **62** via couplings **62-1** and **62-2** formed in the upper part of the holder base **62**. The grips **64-1** and **64-2** are formed to extend from the couplings **62-1** and **62-2** toward a center of the holder base **62**. The grips **64-1** and **64-2** are gripped with an unshown chuck, when the anode holder **60** is conveyed to the plating solution tank **50**.

(48) An electrode terminal **82** for applying a voltage to the anode **40** is disposed in a lower part of

the arm **70-1** extending outward from the couplings **62-1** and **62-2**. The electrode terminal **82** is connected to the positive electrode of the power source **90**, when the anode holder **60** is housed in the plating solution tank. Also, the anode holder **60** includes a power supply member **89** extending from the electrode terminal **82** to a substantially central portion of the inner space **61**. The power supply member **89** is a substantially plate-shaped conductive member, and electrically connected to the electrode terminal **82**.

(49) As shown in FIG. 3, the anode **40** is fixed to a front surface of the power supply member **89** with a fixing member **88** including, for example, a screw and the like. Consequently, the voltage can be applied from the power source **90** to the anode **40** via the electrode terminal **82** and the power supply member **89**.

(50) An annular opening **62a** for changing the anode **40** is formed in a substantially central portion of the holder base **62**, that is, at a position corresponding to the fixing member **88**. The opening **62a** communicates with a back surface side of the inner space **61**, and is covered with a lid **86**. On a back surface side of the holder base **62**, an annular second sealing member **85** including, for example, an O-ring or the like is disposed along the opening **62a**. A gap between the opening **62a** and the lid **86** is sealed with the second sealing member **85**.

(51) The lid **86** is removed when the anode **40** is changed. Specifically, for example, with elapse of useful life of the anode **40**, an operator removes the lid **86**, and removes the fixing member **88** via the opening **62a**. The operator removes the outer edge mask **68** from the diaphragm retainer **69**, and removes the anode **40** from the inner space **61**. Subsequently, the operator houses another anode **40** in the inner space **61**, and fixes the anode **40** to the front surface of the power supply member **89** with the fixing member **88** via the opening **62a**. Lastly, the operator seals the opening **62a** with the lid **86**, and attaches the outer edge mask **68** to the diaphragm retainer **69**.

(52) A weight **87** is attached to a back surface of the holder base **62**. Consequently, the anode holder **60** can be prevented from floating on a surface of water due to buoyancy, when the anode holder **60** is immersed into the plating solution.

(53) As shown in FIG. 5, the anode holder **60** further includes a valve **91** configured to seal the hole **71**, a spring **96** for biasing the valve **91** to close the valve **91**, a shaft **93** for transmitting biasing force of the spring **96** to the valve **91**, a push rod **95** as an operation member that operates the valve **91** to open and close the valve, and an intermediate member **94** for transmitting, to the shaft **93**, force applied to the push rod **95**.

(54) The valve **91** is disposed in the holder base **62** so that the hole **71** can be sealed on an inner side of the holder base **62**. The shaft **93** is disposed along an up-down direction in the holder base **62**. The shaft **93** has one end coupled to the valve **91**, and the other end coupled to the spring **96**. Consequently, the shaft **93** transmits the biasing force of the spring **96** to the valve **91**, and the valve **91** is biased so that the hole **71** is sealed with the valve **91** on the inner side of the holder base **62**.

(55) Thus, the anode holder **60** includes the valve **91** that seals the hole **71**, so that the hole **71** can be sealed, after the anode holder **60** is immersed into the plating solution to fill the inner space **61** with the plating solution. Consequently, if oxygen, hypochlorous acid or monovalent copper is generated in the vicinity of the anode **40**, proceeding of decomposition of the additive can be inhibited, because the outer space and the inner space **61** are divided. Alternatively, in the plating apparatus, the anode holder **60** may be disposed in the plating solution storage tank **52** in a state where a base liquid is put in the plating solution storage tank **52**, the inner space **61** of the anode holder **60** may be filled with the base liquid and then sealed, and a liquid containing the additive may be put in the plating solution storage tank **52** to prepare the plating solution in the outer space. In this case, the inner space **61** of the anode holder **60** does not store the additive, and hence consumption of the additive in the vicinity of the anode **40** can be reduced more. However, the present invention is not limited to this example, and the anode holder **60** may be disposed in the plating solution storage tank **52** in a state where the plating solution containing the additive is put

in the plating solution storage tank **52**, and the inner space **61** of the anode holder **60** may be filled with the plating solution containing the additive and then sealed.

(56) Next, description will be made as to fixing of the diaphragm **66** and the mask **67** in the anode holder **60**. FIG. **6A** is a view schematically showing a mounting structure of the diaphragm **66** and the mask **67** in FIG. **3**. Note that in FIG. **6A** and the subsequent drawing, a closely connecting layer closely connecting the diaphragm **66** and the mask **67** is denoted with reference number **100**.

Further, in FIG. **6A** and the subsequent drawing, the diaphragm **66** is fixed to one of a front surface or a back surface of the mask **67**, but the present invention is not limited to this example, and the diaphragm may be fixed to the other of the front surface or the back surface of the mask **67**.

Alternatively, the diaphragms **66** may be fixed to both of the front surface and the back surface of the mask **67**, and the masks **67** may be fixed to both of front and rear of the diaphragm **66**, respectively.

(57) In an example shown in FIG. **6A**, both the diaphragm **66** and the mask **67** have a size larger than a size of an opening in the diaphragm retainer **69**, and both the diaphragm **66** and the mask **67** are sandwiched between the diaphragm retainer **69** and the holder base cover **63**, to be supported in the anode holder **60**. Also, the diaphragm **66** is closely connected to the mask **67** via the closely connecting layer **100**. According to this configuration, it is possible to tightly close between the anode holder **60** and the diaphragm **66** more securely, and the mask **67** can be physically sandwiched between the diaphragm **66** and the diaphragm retainer **69**, to be firmly supported. In the example shown in FIG. **6A**, the diaphragm **66** is closely connected to the back surface (lower side in FIG. **6**) of the mask **67**. Thus, the mask **67** is located in front of the diaphragm **66**, and hence when oxygen is generated in the inner space **61** where the anode **40** is disposed, oxygen can be inhibited from entering the holes **67a**, and a disadvantage that oxygen is located in the holes **67a** to shield the electric field can be suppressed.

(58) FIG. **6B** is a view schematically showing another example of the mounting structure shown in FIG. **6A**. An example shown in FIG. **6B** is different from the example shown in FIG. **6A** in that instead of closely connecting the diaphragm **66** and the mask **67** via the closely connecting layer **100**, the diaphragm **66** is pressed onto and fixed to the mask **67** by the anode **40**, and the example is the same as the example shown in FIG. **6A** in the other respects. In the example shown in FIG. **6B**, the diaphragm **66** is pressed onto and fixed to the mask **67** from the inner space **61** side by the anode **40**. In other words, the diaphragm **66** is sandwiched between and supported by the mask **67** and the anode **40** in a region that covers the opening **63a** of the holder base **62**. Also, according to this configuration, an effect similar to that of FIG. **6B** can be exhibited. Additionally, the example shown in FIG. **6B** does not include the closely connecting layer **100**, but the present invention is not limited to this example, and the diaphragm **66** may be sandwiched between and supported by the mask **67** and the anode **40**, and closely connected to the mask **67** via the closely connecting layer **100**. Also, in an example of a mounting structure in which the diaphragm **66** and the mask **67** are closely connected via the closely connecting layer **100** as shown in FIGS. **7** to **9** below, the diaphragm **66** may be sandwiched and fixed between the mask **67** and the anode **40** in place of or in addition to the closely connecting layer **100**.

(59) FIG. **7** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a first modification. In an example shown in FIG. **7**, the diaphragm **66** is formed in a size larger than a size of an opening in a diaphragm retainer **69**, and the mask **67** is formed in a size smaller than the size of the opening in the diaphragm retainer **69**. Then, the diaphragm **66** is sandwiched between and supported by the diaphragm retainer **69** and a holder base cover **63**, and the mask **67** is fixed to a front surface (upper side in FIG. **7**) of the diaphragm **66** to be indirectly supported via the diaphragm **66**. Here, in the case of adopting a configuration shown in FIG. **7**, it is presumed that a gap is generated between an outer peripheral end of the mask **67** and the diaphragm retainer **69**, and hence a sealing member **102** may be disposed to seal the gap. According to this configuration, it is possible to tightly close between the anode holder **60** and the

diaphragm **66** more securely.

(60) FIG. **8** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a second modification. In an example shown in FIG. **8**, the diaphragm **66** is formed in a size smaller than a size of an opening in a diaphragm retainer **69**, and the mask **67** is formed in a size larger than the size of the opening in the diaphragm retainer **69**. Then, the mask **67** is sandwiched between and supported by the diaphragm retainer **69** and a holder base cover **63**, and the diaphragm **66** is fixed to a back surface (lower side in FIG. **8**) of the mask **67** to be indirectly supported via the mask **67**. According to this configuration, the mask **67** can be physically sandwiched between and firmly supported by the diaphragm **66** and the diaphragm retainer **69**.

(61) FIG. **9** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a third modification. In an example shown in FIG. **9**, the mounting structure is the same as that shown in FIG. **8** except that a diaphragm retainer **69** is not provided and the mask **67** is directly fixed to a holder base cover **63**. In the example shown in FIG. **9**, the diaphragm **66** is formed in a size smaller than a size of an opening in the holder base cover **63**, and the mask **67** is formed in a size larger than the size of the opening in the holder base cover **63**. Also, the mask **67** includes a thick portion **106** with a large thickness around an outer peripheral edge, and has the thick portion **106** fixed to the holder base cover **63** with a screw. The diaphragm **66** is fixed to a back surface (lower side in FIG. **9**) of the mask **67** to be indirectly supported via the mask **67**. Thus, the thick portion **106** is provided, so that rigidity around the outer peripheral edge can be increased, and the mask **67** can be inhibited from being deformed, for example, in the case of heat welding of the diaphragm **66** and the mask **67**. Note that the thick portion **106** of the mask **67** is formed thickly to protrude on a side (upper side in FIG. **9**) opposite to a surface (lower side in FIG. **9**) of the mask that is fixed to the diaphragm **66**.

(62) FIG. **10** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a fourth modification. In an example shown in FIG. **10**, similarly to the example shown in FIG. **9**, a diaphragm retainer **69** is not provided, and the mask **67** is directly fixed to a holder base cover **63**. In the example shown in FIG. **10**, the diaphragm **66** is formed in a size smaller than a size of an opening in the holder base cover **63**, and the mask **67** is formed in a size larger than the size of the opening in the holder base cover **63**. Also, the mask **67** includes a thick portion **106** with a large thickness around an outer peripheral edge, and has the thick portion **106** fixed to the holder base cover **63** with a screw. The diaphragm **66** is fixed to a front surface (upper side in FIG. **10**) of the mask **67** to be indirectly supported via the mask **67**. Note that in the example shown in FIG. **10**, the thick portion **106** of the mask **67** is formed thickly to protrude on a side (lower side in FIG. **10**) opposite to a surface (upper side in FIG. **10**) of the mask that is fixed to the diaphragm **66**. Thus, the thick portion **106** is provided, so that rigidity around the outer peripheral edge can be increased, and the mask **67** can be inhibited from being deformed, for example, in the case of heat welding of the diaphragm **66** and the mask **67**. Further, the thick portion **106** is formed to protrude rearward, so that a volume of an inner space **61** can be increased. Also, the thick portion **106** has an inner peripheral edge tapered to be smoothly continuous with a region to which the diaphragm **66** is fixed. This can prevent oxygen generated in the inner space **61** from staying in the inner space **61**, and oxygen can be smoothly exhausted from an air outlet **81**. Further, the mask **67** is located behind (on an inner space **61** side of) the diaphragm **66** as shown in FIG. **10**, so that it is possible to reduce the possibility that the fixed diaphragm **66** and mask **67** peel off, when a plating solution in a plating solution storage tank **52** is stirred with a paddle **16**. Note that in the example shown in FIG. **10**, the diaphragm **66** is formed in the size smaller than the size of the opening in the holder base cover **63**, but may be formed in a size larger than the size of the opening in the holder base cover **63**.

(63) FIG. **11** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a fifth modification. In an example shown in FIG. **11**, the mask **67** is formed in a size larger than a size of an opening in a diaphragm retainer **69**, and sandwiched between the

diaphragm retainer **69** and a holder base cover **63** to support the mask **67**. On the other hand, in the example shown in FIG. **11**, a plurality of diaphragms **66** are provided in a shape corresponding to each of a plurality of holes **67a** of the mask **67**. Then, the plurality of diaphragms **66** are fixed to the mask **67** to cover each of the plurality of holes **67a** of the mask **67**, and are accordingly indirectly supported via the mask **67**. Here, in the example shown in FIG. **11**, the plurality of holes **67a** in the mask **67** include stepped portions **67b** each formed in a size smaller than a size of the diaphragm **66**, and the diaphragms **66** are fixed to the stepped portions **67b**, respectively, to fix the diaphragms **66** and the mask **67**. Further, as shown in FIG. **11**, to more securely seal the mask **67** and each diaphragm **66**, a circular sealing member **104** to be bonded or welded to at least one of the mask **67** and the diaphragm **66** may be provided. In this case, the diaphragm **66** and the mask **67** may be closely connected to each other via a closely connecting layer **100**, or may be fixed to each other via no closely connecting layer **100** and via the sealing member **104**. Also, in this case, a region where the diaphragm **66** is in contact with a plating solution can be reduced, and consumption of an additive can be reduced.

(64) FIG. **12** is a view schematically showing a mounting structure of a diaphragm **66** and a mask **67** according to a sixth modification. The mounting structure shown in FIG. **12** is the same as the mounting structure shown in FIG. **7** except a fixing method of the diaphragm **66** and the mask **67**. In the example shown in FIG. **12**, the mask **67** and the diaphragm **66** are fixed to each other with a screw via no closely connecting layer **100**. Here, the mask **67** and the diaphragm **66** are fixed with the screw in openings in a holder base cover **63** and a diaphragm retainer **69**. In other words, the diaphragm **66** and the mask **67** are not fixed to each other in a region (first region) to be sandwiched between the holder base cover (base body) **63** and the diaphragm retainer **69**, but are fixed to each other in a region (second region) that is not supported by the holder base cover **63** and the diaphragm retainer **69**. However, the diaphragm **66** and the mask **67** are not limited to those fixed to each other only in the second region, and may be fixed to each other in the first region. Specifically, in the example shown in FIG. **12**, the mask **67** includes a first mask member **111** disposed in front of (on an upper side in FIG. **12**) the diaphragm **66**, and a second mask member **112** disposed behind (on a lower side in FIG. **12**) the diaphragm **66**. Then, the first mask member **111** and the second mask member **112** between which the diaphragm **66** is sandwiched are fixed with a screw, to fix the mask **67** and the diaphragm **66**. Also, in this example, a region where the diaphragm **66** is in contact with a plating solution can be reduced, and consumption of an additive can be reduced in the same manner as in the other examples.

(65) Next, description will be made as to a specific example of the fixing of the diaphragm **66** and the mask **67**. Each of FIGS. **13** to **21** is a view schematically showing a fixed part between the diaphragm **66** and the mask **67**, and shows, with hatching, a region where the diaphragm **66** and the mask **67** are non-removably fixed. Note that in examples shown in FIGS. **13** to **21**, the diaphragm **66** and the mask **67** are non-removably fixed to each other in a partial region, but part of the diaphragm **66** may only be fixed to the mask **67** in a region that covers a front surface of an inner space **61**, that is, a region that covers an opening **63a** of a holder base cover **63**, or the diaphragm and the mask may be non-removably fixed to each other in the whole region. In addition, when non-removably fixing the diaphragm **66** and the mask **67**, welding, bonding or the like may be used as described above. Further, in the examples shown in FIGS. **13** to **21**, a plurality of holes **67a** of the mask **67** are provided at equal intervals in each of a first alignment direction (up-down direction in the drawing) and a second alignment direction (right-left direction in the drawing). Further, in FIGS. **13** to **21**, the up-down direction in the drawing is the same as the up-down direction (vertical direction) in FIG. **1**, but the present invention is not limited to this example, and the direction may be tilted from the up-down direction (vertical direction) in FIG. **1**. Further, in FIGS. **13** to **21**, for ease of description, the diaphragm **66** and the mask **67** have the same outer shape size, but the present invention is not limited to this example.

(66) FIGS. **13** to **16** show a non-removable fixed part between the diaphragm **66** and the mask **67**

according to first to fourth examples. In the first to fourth examples, outer peripheral edges of the diaphragm **66** and the mask **67** are not directly fixed, and the diaphragm and the mask are non-removably fixed to each other in a partial region of an inner peripheral region. It is considered that these examples are especially effective in such a configuration as shown in FIG. **6** where the outer peripheral edges of both the diaphragm **66** and the mask **67** are sandwiched between and supported by the holder base cover **63** and the diaphragm retainer **69**.

(67) Specifically, in the first example shown in FIG. **13**, the diaphragm **66** and the mask **67** are non-removably closely connected in a plurality of closely connecting regions **120** along the first alignment direction (up-down direction in FIG. **13**) of the plurality of holes **67a**. Note that in the example shown in FIG. **13**, the holes **67a** and the closely connecting regions **120** are alternately arranged in the second alignment direction (right-left direction in the drawing), but the present invention is not limited to this example. For example, each closely connecting region **120** along the first alignment direction may be provided for every two or more holes **67a** in the second alignment direction. Note that in the first example shown in FIG. **13**, each closely connecting region **120** may have a long shape in a vertical or horizontal direction as the first alignment direction, or a long shape tilted in the vertical or horizontal direction.

(68) In the second example shown in FIG. **14**, the diaphragm **66** and the mask **67** are non-removably closely connected in a lattice-shaped closely connecting region **120** along each of the first alignment direction (up-down direction in FIG. **14**) and the second alignment direction (right-left direction in FIG. **14**) of the plurality of holes **67a**. Note that in the example shown in FIG. **14**, each closely connecting region **120** is disposed for every two holes **67a** in each of the first alignment direction and the second alignment direction, but the present invention is not limited to this example. For example, the closely connecting region **120** may be provided for each hole **67a** or every three or more holes **67a** in the first alignment direction or the second alignment direction. Furthermore, the closely connecting regions **120** may be provided at intervals that are different between the first alignment direction and the second alignment direction.

(69) In the third example shown in FIG. **15**, the diaphragm **66** and the mask **67** are non-removably closely connected in a closely connecting region **120** including a plurality of small regions. In other words, the diaphragm **66** and the mask **67** are closely connected to a plurality of small closely connecting points. Note that in the example shown in FIG. **15**, the closely connecting region **120** is disposed for every two holes **67a** in each of the first alignment direction and the second alignment direction, but the present invention is not limited to this example. For example, the closely connecting region **120** may be provided for each hole **67a** or every three or more holes **67a** in the first alignment direction or the second alignment direction. Furthermore, the closely connecting regions **120** may be provided at intervals that are different between the first alignment direction and the second alignment direction.

(70) In the fourth example shown in FIG. **16**, the diaphragm **66** and the mask **67** are non-removably closely connected in edge portions of the plurality of holes **67a**. Note that in the example shown in FIG. **16**, the edge portions of all the plurality of holes **67a** are formed as closely connecting regions **120**, but the edge portions of some holes **67a** of the plurality of holes **67a** may be formed as the closely connecting regions **120**.

(71) FIGS. **17** to **21** show a fixed part between the diaphragm **66** and the mask **67** according to fifth to ninth examples. In the fifth to ninth examples, outer peripheral edges of the diaphragm **66** and the mask **67** are non-removably closely connected by closely connecting regions **120**. It is considered that these examples are especially effective in configurations shown in FIGS. **7** to **10** where the outer peripheral edge of at least one of the diaphragm **66** and the mask **67** is not sandwiched between the holder base cover **63** and the diaphragm retainer **69**.

(72) Specifically, in the fifth example shown in FIG. **17**, the diaphragm **66** and the mask **67** are non-removably fixed in the outer peripheral edge of the diaphragm **66** or the mask **67**, and are not directly fixed non-removably in an inner peripheral region. Furthermore, the sixth to ninth

examples shown in FIGS. **18** to **21** are the same as the first to fourth examples shown in FIGS. **13** to **16** except that the diaphragm **66** and the mask **67** are non-removably fixed in the outer peripheral edge of the diaphragm or the mask. Redundant description with reference to FIGS. **18** to **21** will not be repeated.

Second Embodiment

(73) FIG. **22** is a schematic view showing a plating apparatus according to a second embodiment. The plating apparatus according to the second embodiment is different from the plating apparatus according to the first embodiment in that a diaphragm **66** and a mask **67** are not attached to an anode holder **60**, but are mounted in an opening **14a** in a regulation plate **14**. In the following description, a description that overlaps with that of the first embodiment will not be repeated.

(74) In the plating apparatus according to the second embodiment, a shield box **160** is disposed in a plating solution storage tank **52**, and accordingly, an interior of the plating solution storage tank **52** is divided into an anode tank **170** inside the shield box **160** and a cathode tank **172** outside the shield box. In the example shown in FIG. **22**, the anode holder **60** holding an anode **40** and the regulation plate **14** are arranged in the anode tank **170**, and a paddle **16** and a substrate holder **18** (cathode) are arranged in the cathode tank **172**.

(75) The shield box **160** includes an opening **160a** at a position corresponding to the opening **14a** of the regulation plate **14**. Also, a tubular part that defines the opening **14a** of the regulation plate **14** is fitted into the opening **160a** of the shield box **160**. According to this configuration, the anode tank **170** communicates with the cathode tank **172** through the opening **14a** of the regulation plate **14**. Then, in the second embodiment, the diaphragm **66** and the mask **67** are mounted in the opening **14a** of the regulation plate **14**, and the anode tank **170** and the cathode tank **172** are divided by the diaphragm **66** and the mask **67**. Alternatively, the diaphragm **66** and the mask **67** may be mounted from an anode tank **170** side in the regulation plate **14**, or may be mounted from a cathode tank **172** side.

(76) As an example, the diaphragm **66** and the mask **67** are mounted to the regulation plate **14** by use of an annular diaphragm retainer **69**. Here, the diaphragm **66** and the mask **67** may be fixed in the regulation plate **14** in the same manner as in fixing the diaphragm **66** and the mask **67** in the anode holder **60** in the first embodiment. That is, as an example, the diaphragm **66** and the mask **67** may be mounted to the regulation plate **14** with a mounting structure in which the holder base cover **63** in the mounting structure shown in FIGS. **6** to **12** is replaced with the regulation plate **14**. Also, the diaphragm **66** and the mask **67** may be fixed in the same manner as in the first embodiment.

(77) In the plating apparatus of the second embodiment, a plating solution in the cathode tank **172** flows over a side wall of the plating solution storage tank **52** to flow into an overflow tank **54**. On the other hand, the plating solution in the anode tank **170** is configured not to overflow. Further, a liquid discharge line **190** in which an on-off valve **186** is disposed is connected to the anode tank **170**. For example, in a case where a soluble anode is used as an anode **40**, a black film generated in the anode tank **170** can be discharged to outside through the liquid discharge line **190**. Therefore, according to the plating apparatus of the second embodiment, an amount of the black film to be included in the plating solution (base liquid) in the anode tank **170** can be decreased, and the black film floating in the plating solution can be substantially completely inhibited from entering the cathode tank **172**.

(78) Also, in the plating apparatus according to the second embodiment, a base liquid supply line **158** is connected to a plating solution circulation line **58a**. The base liquid supply line **158** is not intended to supply the plating solution to the plating solution storage tank **52** during plating of a substrate W, but is used to first supply the base liquid to the plating solution storage tank **52** for performing plating, that is, used only for so-called initial make-up of an electrolytic bath. The base liquid supply line **158** is provided with a first supply valve **151**. Also, in the plating apparatus of the second embodiment, a connection line **192** is disposed to connect the plating solution circulation

line **58a** and the liquid discharge line **190**. The connection line **192** is provided with a second supply valve **152**. Further, the plating apparatus of the second embodiment is provided with an additive supply line **159** for supplying an additive to the cathode tank **172**. The additive supply line **159** is provided with a third supply valve **153**. Usually, the first to third supply valves **151** to **153** are closed.

(79) According to the plating apparatus of the second embodiment, the first supply valve **151** and the second supply valve **152** are opened only during the initial make-up of the electrolytic bath, and the base liquid from the base liquid supply line **158** is supplied through the liquid discharge line **190** and the plating solution circulation line **58a** into the anode tank **170** and the cathode tank **172**. Then, the third supply valve **153** is opened, to supply the additive only to the cathode tank **172**. According to this configuration, the anode tank **170** does not store the additive, and hence consumption of the additive in the vicinity of the anode **40** can be reduced.

(80) In the plating apparatus of the second embodiment described above, the plating solution storage tank **52** is divided into the anode tank **170** and the cathode tank **172** by the shield box **160** and the regulation plate **14**. Then, the diaphragm **66** and the mask **67** including a plurality of holes and fixed to the diaphragm **66** are provided in the opening **14a** of the regulation plate **14**. According to this configuration, a region where the diaphragm **66** is in contact with the plating solution can be reduced, and the additive can be inhibited from reaching the anode **40** to reduce consumption of the additive, in the same manner as in the plating apparatus of the first embodiment.

Third Embodiment

(81) FIG. **23** is a schematic view showing a plating apparatus according to a third embodiment. The plating apparatus according to the third embodiment includes a shield box **160**, and a diaphragm **66** and a mask **67** are mounted in an opening **14a** in a regulation plate **14**, in the same manner as in the second embodiment. The plating apparatus according to the third embodiment is different from the plating apparatus according to the second embodiment in a configuration concerning a plating solution storage tank **52** and the shield box **160**, and is the same as the plating apparatus according to the second embodiment in the other respects. In the following description, a description that overlaps with that of the second embodiment will not be repeated.

(82) In the plating apparatus according to the third embodiment, a bottom plate **51** is disposed in a plating solution storage tank **52**, and an interior of the plating solution storage tank **52** is divided, by this bottom plate, into an upper substrate treatment chamber and a lower plating solution distributing chamber **53**. The shield box **160** is disposed in the upper substrate treatment chamber. The substrate treatment chamber is divided into an anode tank **170** and a cathode tank **172** by the shield box **160**.

(83) The plating apparatus of the third embodiment is configured so that a plating solution in the cathode tank **172** can overflow to flow into an overflow tank **54** and the plating solution in the anode tank **170** does not overflow, in the same manner as in the plating apparatus of the second embodiment. One end of a plating solution circulation line **58a** is connected to a bottom of the overflow tank **54**, and the other end of the plating solution circulation line **58a** is connected to a bottom of the plating solution distributing chamber **53**.

(84) A shielding plate **51c** hanging downward to regulate flow of the plating solution is attached to the bottom plate **51** in the plating solution storage tank **52**. Also, in the bottom plate **51**, a first plating solution flow port **51a** that communicates between the cathode tank **172** and the plating solution distributing chamber **53** is formed. Furthermore, a second plating solution flow port **51b** located below the anode tank **170** is formed in the bottom plate **51**. In a bottom of the shield box **160**, a bottom opening is formed at a position corresponding to the second plating solution flow port **51b**. The plating solution distributing chamber **53** communicates with the anode tank **170** through the second plating solution flow port **51b** and the bottom opening of the shield box **160**. The bottom opening of the shield box **160** is usually sealed with a plating solution plug **210**. The

plating solution plug **210** is connected to a plating solution unplugging stick **212** extending in an up-down direction to outside the shield box **160**. The plating solution unplugging stick **212** moves in a vertical direction, to open and close an opening **160b**. Here, the plating solution unplugging stick **212** may be manually operated, or may be operated by any power source such as a motor, a solenoid, or a pneumatic actuator.

(85) In the plating apparatus of the third embodiment, the plating solution containing an additive is stored in the plating solution storage tank **52** during initial make-up of the electrolytic bath.

Subsequently, the shield box **160** is placed in the plating solution in a state where the plating solution plug **210** is opened, and the anode tank **170** is filled with the plating solution. Then, the plating solution plug **210** is closed, to divide the anode tank **170** and the cathode tank **172**.

(86) Also, in the plating apparatus of the third embodiment, the substrate treatment chamber is divided into the anode tank **170** and the cathode tank **172** by the shield box **160** and the regulation plate **14**. Then, the diaphragm **66** and the mask **67** including a plurality of holes and fixed to the diaphragm **66** are mounted in the opening **14a** of the regulation plate **14**. Consequently, a region where the diaphragm **66** is in contact with the plating solution can be reduced, and the additive in the cathode tank **172** can be inhibited from reaching an anode **40** to reduce consumption of the additive, in the same manner as in the plating apparatus of the first embodiment.

(87) Note that in the first to third embodiments, the diaphragm **66** and the mask **67** are arranged to extend in a vertical direction of the plating apparatus (to orient plate surfaces in a horizontal direction), but the present invention is not limited to the examples. For example, the diaphragm **66** and the mask **67** may be arranged to extend in the horizontal direction of the plating apparatus (to orient the plate surfaces in the vertical direction).

(88) The present embodiments described above can be described in aspects as follows.

(89) [Aspect 1]

(90) According to Aspect 1, an anode holder for holding an anode for use in a plating apparatus is provided. The anode holder includes an inner space formed in the anode holder, to house the anode, a mask including a plurality of holes, and configured to cover a front surface of the inner space, and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers the front surface of the inner space. According to the anode holder of Aspect 1, the mask can reduce a region where the diaphragm is in contact with a plating solution, and can further inhibit an additive from reaching the anode to reduce consumption of the additive.

(91) [Aspect 2]

(92) According to Aspect 2, in Aspect 1, the diaphragm and the mask are closely connected to each other via a closely connecting layer.

(93) [Aspect 3]

(94) According to Aspect 3, in Aspect 1 or 2, the diaphragm and the mask are bonded or welded to each other.

(95) [Aspect 4]

(96) According to Aspect 4, an opening ratio by the plurality of holes is equal to or more than 2% and equal to or less than 25%.

(97) Aspect 5

(98) According to Aspect 5, in Aspects 1 to 4, the anode holder includes a base body supporting at least one of the diaphragm and the mask, and the diaphragm and the mask are fixed to each other in a second region that is different from a first region where the at least one of the diaphragm and the mask is supported by the base body. According to Aspect 5, the plating solution can be inhibited from entering a gap between the diaphragm and the mask, and the consumption of the additive can be further reduced.

(99) [Aspect 6]

(100) According to Aspect 6, in Aspects 1 to 5, the mask is fixed on a side of the inner space with respect to the diaphragm.

(101) [Aspect 7]

(102) According to Aspect 7, in Aspects 1 to 5, the mask is fixed on a side opposite to the inner space with respect to the diaphragm.

(103) [Aspect 8]

(104) According to Aspect 8, in Aspect 7, the diaphragm is sandwiched between the mask and the anode to be fixed to the mask.

(105) [Aspect 9]

(106) According to Aspect 9, in Aspects 1 to 8, each of the plurality of holes is tapered with a diameter increasing away from the diaphragm. According to Aspect 9, foreign matter can be inhibited from staying in the plurality of holes in the mask.

(107) [Aspect 10]

(108) According to Aspect 10, in Aspects 1 to 9, the diaphragm and the mask are arranged to extend in a vertical direction of the plating apparatus.

(109) [Aspect 11]

(110) According to Aspect 11, in Aspects 1 to 10, the mask is made of a resin.

(111) [Aspect 12]

(112) According to Aspect 12, in Aspects 1 to 11, the diaphragm is an ion exchange membrane or a neutral diaphragm.

(113) [Aspect 13]

(114) According to Aspect 13, a plating apparatus is provided. The plating apparatus includes a plating solution tank, a mask including a plurality of holes, and dividing the plating solution tank into an anode tank in which an anode is disposed and a cathode tank in which a cathode is disposed, and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers a front surface of an inner space. According to the plating apparatus of Aspect 13, the mask can reduce a region where the diaphragm is in contact with a plating solution, and can further inhibit an additive from reaching the anode to reduce consumption of the additive.

(115) The embodiments of the present invention have been described above based on several examples, but the above embodiments of the present invention are described to facilitate understanding of the present invention, and do not limit the present invention. The present invention may be changed or modified without departing from the scope, and needless to say, the present invention includes equivalents to the invention. Also, in a range in which at least some of the above-described problems can be solved or a range in which at least some of effects are exhibited, any arbitrary combination or omission of respective constituent components described in claims and description is possible.

(116) The present application is based on and claims the benefit of priority of Japanese Patent Application No. 2019-107724 filed on Jun. 10, 2019. All disclosed contents including the description, claims, drawings and abstract of Japanese Patent Application No. 2019-107724 are entirely incorporated herein by reference. All disclosure including the description, claims, drawings and abstract of each of Japanese Patent No. 2510422 (PTL 1) and Japanese Patent Laid-Open No. 2009-155726 (PTL 2) is entirely incorporated herein by reference.

REFERENCE SIGNS LIST

(117) **14** regulation plate **14a** opening **16** paddle **18** substrate holder **40** anode **50** plating solution tank **52** plating solution storage tank **54** overflow tank **60** anode holder **61** inner space **62** holder base **63** holder base cover **66** diaphragm **67** mask **67a** hole **68** outer edge mask **69** diaphragm retainer **100** closely connecting layer **102** sealing member **104** sealing member **106** thick portion **108** shield box **111** first mask member **112** second mask member **120** closely connecting region **160** shield box **170** anode tank **172** cathode tank

Claims

1. An anode holder for holding an insoluble anode for use in a plating apparatus, the anode holder comprising: an inner space formed in the anode holder, to house the insoluble anode; a mask comprising a plate-shaped member including a plurality of holes, and configured to cover a front surface of the inner space; and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers the front surface of the inner space, wherein the inner space and an outer space of the anode holder are divided by the diaphragm and the mask, the mask is fixed on a side opposite to the inner space with respect to the diaphragm, and the diaphragm is sandwiched between the mask and the insoluble anode to be fixed to the mask, and is placed in direct physical contact with the insoluble anode.
 2. The anode holder according to claim 1, wherein the diaphragm and the mask are closely connected to each other via a closely connecting layer.
 3. The anode holder according to claim 1, wherein the diaphragm and the mask are bonded or welded to each other.
 4. The anode holder according to claim 1, wherein an opening ratio by the plurality of holes is equal to or more than 2% and equal to or less than 25%.
 5. The anode holder according to claim 1, further comprising a base body supporting at least one of the diaphragm and the mask, wherein the diaphragm and the mask are fixed to each other in a second region that is different from a first region where the at least one of the diaphragm and the mask is supported by the base body.
 6. The anode holder according to claim 1, wherein each of the plurality of holes is tapered with a diameter increasing away from the diaphragm.
 7. The anode holder according to claim 1, wherein the diaphragm and the mask are arranged to extend in a vertical direction of the plating apparatus.
 8. The anode holder according to claim 1, wherein the mask is made of a resin.
 9. The anode holder according to claim 1, wherein the diaphragm is an ion exchange membrane or a neutral diaphragm.
 10. A plating apparatus comprising: a plating solution tank; a mask comprising a plate-shaped member including a plurality of holes, and dividing the plating solution tank into an anode tank in which an insoluble anode is disposed and a cathode tank in which a cathode is disposed; and a diaphragm, at least part of the diaphragm being fixed to the mask in a region of the mask that covers a front surface of the anode tank, wherein the anode tank and the cathode tank are divided by the diaphragm and the mask, the mask is fixed on a side of the cathode tank with respect to the diaphragm, and the diaphragm is sandwiched between the mask and the insoluble anode to be fixed to the mask, and is placed in direct physical contact with the insoluble anode.
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